ELECTRIC PROPERTIES OF MACROMOLECULES. V. THEORY OF IONIC POLARIZATION IN POLYELECTROLYTES

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Ionic transport phenomena are shown to be important to the dielectric constant, as well as the dispersion and conductivity of polyelectrolyte solutions. The effects of the excess conductivity arising from mobility of ions at the interface, and—with thin ion atmospheres—the charge transport due to counterions, may be expressed in terms of a two-dimensional conductivity. The electrical boundary value problem is formulated in terms of this surface conductivity, the dielectric constant and volume conductivity of the medium, and an anisotropic dielectric constant and volume conductivity for the particle. These parameters have important physical significance for macromolecular structures. It is proposed that this model is more appropriate for polyelectrolytes than either the Debye–Falkenhagen or Maxwell–Wagner models. Isotropic spheres are considered in detail, to show the effects of a surface conductivity on the internal field, the induced polarization, and the dielectric constant, dispersion, and conductivity of dilute systems. Reasonable approximations were found which obviate detailed treatment of the boundary-value problems for the generalized ellipsoid and the limiting forms of a cylinder. Anisometric particles of uniform surface conductivity are shown to be electrically equivalent to particles of anisotropic volume conductivity. Equations are given for the complex dielectric constant, the relaxation times, and the low and high frequency dielectric and conductivity increments of dilute systems of ellipsoidal particles. Numerical calculations of the increments are made for randomly and completely oriented ellipsoids of revolution over a wide range of parameters appropriate to dilute aqueous media. Important differences between needle and disk-shaped particles are found. In oriented systems of anisometric particles, counterion transport produces a strong anisotropy of the low frequency dielectric increment, and an anisotropy of conductivity apart from the contribution due to anisotropy of the frictional coefficient of the polyion. Further, the internal field depends upon orientation, and this effect on polyion mobility may greatly exceed the frictional coefficient effect discussed by Eigen and Schwarz. The results indicate that essentially all of the known dielectric properties of aqueous proteins, nucleic acids, nucleoproteins, and charged colloids are explicable in terms of this model. It is shown that polarization of the counterion atmosphere reduces the internal field and thus diminishes permanent dipole polarization. Available experimental data, generally incomplete for the present interpretation, are discussed in as far as possible. The need for further experiments to test the theory is indicated, and some crucial ones are suggested. It is predicted that biological effects in intense high frequency fields may be enhanced by the use of pulsed radiation.

1. Introduction

Of the many types of systems for which the frequency-dependent dielectric properties have been investigated, perhaps the least understood are solutions of polyelectrolytes and colloidal electrolytes. These classes include the proteins, nucleic acids, synthetic polyelectrolytes, micelles and many colloidal dispersions. Formulation of an adequate theory of the dielectric constant and conductivity of these systems requires treatment of the charge transport processes arising from the motion of the ions or other charge carriers in the solvent, in the ion atmosphere, and in the solutes or the suspended phase. Neglect of these effects can lead to serious errors in the interpretation of dielectric constant and conductivity. A recent interpretation evidently in error will be discussed below.

There exists no general theory of dielectric constant and conductivity rigorously applicable to polyelectrolytes. Debye and Falkenhagen[1] treated the dielectric constant of simple electrolytes in terms of a model involving point charges in a dielectric continuum. Several investigators[2] have treated the conductivity of simple electrolytes and in recent years there have been some significant refinements[2b,3] which are in accord with experimental results well above the low concentrations for which the Debye–Hückel limiting law is valid. But the refinements are not adequate for polyelectrolytes, where particle dimensions and shape are important constants of the system.

A central concept in the treatment of simple electrolytes is the existence of an ion atmosphere. In external fields, the ion atmosphere becomes polarized; by this we refer to the production of a dipole moment by displacement of the center of atmospheric charge from the center of the charge about which the atmosphere is formed. This results in a retarding local field which depresses the equivalent conductivity of an ion below the infinite dilution value, often referred to as the "asymmetry effect" or the "relaxation effect." For highly charged polyelectrolytes with many neighboring counterions, and a greater distance for them to move, we may anticipate that this effect will be larger than in simple electrolytes. To treat it, we wish to adopt a tractable model in which mobilities enter naturally, and which depicts the sharp gradient of ion concentrations in the strong electric field of the polyelectrolyte. It was proposed recently[4] that the introduction of a surface conductivity in treating the electrical boundary-value problems of polyelectrolyte systems may be helpful in explaining some of the interesting dielectric behavior of protein solutions. This model is treated mathematically here.

The concept of a surface conductivity apparently was first introduced by Smoluchowski,[5] who recognized the importance of the mobility of the ions of the diffuse layer at a charged surface, and employed this concept in discussing electrophoresis of colloidal particles. The idea subsequently was elaborated by Havestadt and R. Fricke,[6] Bikerman,[7] H. Fricke and Curtis[8] and Henry.[9] In fact,

(1) (a) P. Debye and H. Falkenhagen, *Physik. Z.*, **29**, 121, 401 (1928); (b) *Z. Elektrochem.*, **34**, 562 (1928).

(2) (a) L. Onsager, *Physik. Z.*, **27**, 388 (1926); **28**, 277 (1927); (b) R. M. Fuoss and L. Onsager, *Proc. Natl. Acad. Sci.*, **41**, 274, 1010 (1955); THIS JOURNAL, **61**, 668 (1957).

(3) (a) R. M. Fuoss, *J. Am. Chem. Soc.*, **81**, 2658 (1959); (b) H. Falkenhagen, M. Leist and G. Kelbg, *Ann. Physik*, [6] **11**, 51 (1952).

(4) C. T. O'Konski, *J. Chem. Phys.*, **23**, 1559 (1955).

(5) M. von Smoluchowski, "Electric Endosmosis and Streaming Current," in "Two Monographs on Electrokinetics," trans. by P. E. Bocquet [Ann Arbor] Engineering Research Institute, University of Michigan, 1951, pp. 51–159.

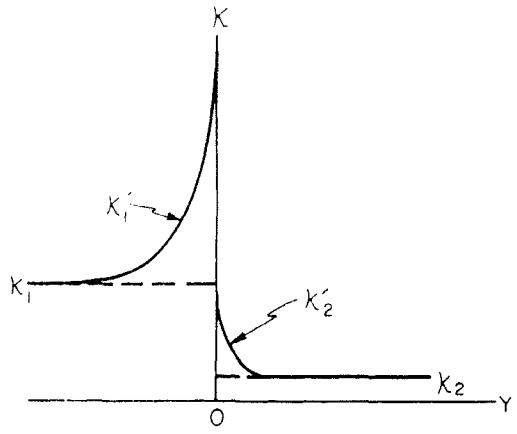


Fig. 1.—Physical situation prompting introduction of a surface conductivity. The primed quantities show the conductivity variation in the vicinity of a planar charged "interface." A contribution from an infinitely thin layer, or a "true" surface conductivity, would correspond to a singularity at $y = 0$.

Miles and Robertson[10] proposed very early that the electric properties of colloidal particles with an electric double layer might be explained in terms of a model involving a sphere surrounded by a concentric shell of higher conductivity, and developed equations for the complex conductivity of a dilute system. But the parameters were not evaluated and the dielectric increment was not discussed. Bikerman[7b] obtained an approximate expression for the dielectric constant of a dilute suspension of cylindrical rods with surface conductivity. Subsequently, the theory of Maxwell–Wagner polarization was extended to homogeneous ellipsoids by Fricke,[11] but a surface conductivity was not incorporated in the model.

In the present study, the concept of a surface conductivity is generalized to include contributions from all species in the ion atmosphere, and other charge carriers, such as protons, electrons or holes, which are confined to a thin region on or near the surface. Equations are then obtained for the dielectric constant and conductivity of dilute systems of variously shaped particles of arbitrary dielectric constant, volume conductivity and surface conductivity, completely oriented or randomly oriented, in a solvent of arbitrary dielectric constant and conductivity, at arbitrary frequency. Various special cases are examined and predictions are made for some systems of particular interest. It is shown that remarkably high dielectric increments, conductivity increments and a broad range of relaxation behavior can be explained in terms of this model. It is concluded that the model has a useful range of validity in interpreting the behavior of solutions of large rigid macromolecules and colloidal electrolytes, and that it provides a system of equations to facilitate further investigation of mobile charges in the ion atmosphere and on surfaces, through dielectric constant and conductivity studies.

2. The Surface Conductivity Model and Ion Atmosphere Polarization

The surface conductivity may be defined by means of the relation

$$dq_s/dt = \lambda E_t l \qquad (2.1)$$

where the first term is the rate of transfer of charge along a surface of specific conductivity λ through a line element of length l in an electric field with a component E_t along the surface and perpendicular to the line element. The charge carriers may be ions, electrons or holes. If u_i is the mobility of a carrier of type i, then the contribution to λ from all carriers in the surface is

$$\lambda_s = \sum_i s_i u_i z_i \qquad (2.2)$$

where s_i is the number of carriers per unit surface area, and z_i is the absolute value of the valence.

In the more general case, carriers will not necessarily be restricted to a surface, but may be transported through the bulk media as well. Furthermore, their concentrations will be functions of position with respect to the surface. When the region of conductivity variation is thin compared to the dimensions of the particles the local conductivity at each point may be regarded as the sum of a bulk value, *i.e.*, a value characteristic of the phase at a specified distance from the surface, and a value to be associated with the surface. Then the surface conductivity may be written as the sum of λ_s, defined above, and λ_a, the contribution to be evaluated from the excess conductivity due to the ion atmosphere. Consider, for example, a plane interface between the two regions illustrated in Fig. 1. Let the bulk conductivities of regions 1 and 2 at large distances from the surfaces be κ_1 and κ_2, and the conductivities at arbitrary distances by κ_1' and κ_2', respectively. Then the total surface conductivity is

$$\lambda = \lambda_s + \lambda_a = \sum_i s_i u_i z_i + \int_{-\infty}^{0} (\kappa_1' - \kappa_1)dy + \int_0^{\infty} (\kappa_2' - \kappa_2)dy \qquad (2.3)$$

For dilute systems, we may compute the bulk conductivities of each phase from the individual ionic contributions in the usual manner

$$\kappa_k' = \frac{1}{N} \sum_i z_i n_i(y)\, l_i(y) \qquad k = 1,2 \qquad (2.4)$$

where n_i is the number of ions of type i per cc. and $l_i(y)$ is the equivalent conductance of species i at position y. By considering that the equivalent conductivity is a function of distance, we allow for variations in properties of ions and solvent in the strong fields prevailing near polyions.

Calculations of the surface conductivity have been made on various double-layer models, and it already has been suggested that studies of surface conductance may permit a definitive evaluation of the merits of the Helmholtz, Gouy and Stern treat-

(6) (a) L. Havestadt and R. Fricke, *Z. anorg. allgem. Chem.*, **188**, 357 (1930); **196**, 120 (1931); (b) R. Fricke, *Kolloid Z.*, **56**, 166 (1931).

(7) (a) J. J. Bikerman, *Z. physik. Chem.*, **A171**, 209 (1934); (b) *J. chim. phys.*, **32**, 285 (1935); (c) *Kolloid-Z.*, **72**, 100 (1935); (d) *Trans. Faraday Soc.*, **36**, 154 (1940).

(8) H. Fricke and H. J. Curtis, THIS JOURNAL, **41**, 729 (1937).

(9) D. C. Henry, *Trans. Faraday Soc.*, **44**, 1021 (1948).

(10) J. B. Miles and H. P. Robertson, *Phys. Rev.*, **40**, 583 (1932).

(11) H. Fricke, THIS JOURNAL, **57**, 934 (1953).

ments.[12] It is probable that adsorbed or bound ions may move along the surface. A particular example of this has been discussed, namely, the possibility of proton transfer on the surface of proteins in solution.[13] The importance of this effect has not been demonstrated conclusively, but the term λ_s is general and may include a contribution from it. The rest of the expression for λ refers to the ion atmospheres outside and inside the particle in question. In many cases, the internal ion atmosphere contribution may be negligible owing to low ionic concentrations and mobilities.

3. The Surface Conductivity Effect for Spheres in a Conducting Medium in a Periodic Field

We consider uniform spheres of radius a, dielectric constant ϵ_2, bulk conductivity κ_2, and surface conductivity λ in a solvent of dielectric constant ϵ_1 and conductivity κ_1, under the influence of an external electric field. The first step is to find expressions for the potentials within and outside a given sphere. In setting up the boundary value problem, we adopt the method of Böttcher[14] and consider that each sphere is surrounded by a homogeneous medium of the bulk properties of the system. This is a useful approximation for the average effect of the surrounding particles at moderate concentrations. However, it does not account for the effects of local fluctuations in the distribution of neighboring particles. We introduce polar coördinates r, θ, ϕ, with origin at the center of the sphere, and z as polar axis. Let the instantaneous value of the macroscopic field intensity be E, along the z direction. Because of the symmetry, everything will be independent of ϕ. At large distances from the sphere the electric potential will be

$$\psi = -Er\cos\theta \tag{3.1}$$

ψ is the component of the total potential arising from the applied field. The quantities ψ and E used without subscripts refer to average values over the system. Joule heating and electrochemical effects are assumed negligible; it is well known that these become unimportant at sufficiently low fields and high frequencies. Both the sphere and the surroundings are considered to have uniform bulk conductivities and dielectric constants, so application of a field does not produce space charge within either region. Therefore we may employ Laplace's equation in both regions

$$\nabla^2\psi_2 = 0;\ \nabla^2\psi = 0 \tag{3.2}$$

At the surface of the sphere we have the usual conditions

$$\psi_2|_a = \psi|_a \tag{3.3}$$

and

$$\epsilon_2 \frac{\partial\psi_2}{\partial r}\Big|_a - \epsilon \frac{\partial\psi}{\partial r}\Big|_a = 4\pi\sigma \tag{3.4}$$

It is important to realize that equation 3.4, which is obtained by application of Gauss's theorem, is valid at any instant. The quantity σ is the density of free charge at the surface, and ϵ_2 and ϵ are the real parts of the complex dielectric constants of sphere and suspension, respectively. Since ψ_2 and ψ are complex where the field is periodic, σ will also be complex.

To solve the problem for a periodic field, given by

$$E = E_0 e^{i\omega t} \tag{3.5}$$

we require a differential equation for σ. Here E_0 is the space mean value of the peak field intensity in the suspension, ω is the angular frequency, and $i = \sqrt{-1}$. The surface charge density undergoes a time variation as the result of two processes: (a) the transport of ions to and from an element of surface as a result of the bulk conductivities of the two media, and (b) the transport of ions along the surface as a result of the surface conductivity. For isotropic media the first contribution is

$$\frac{d\sigma_a}{dt} = \kappa \frac{\partial\psi}{\partial r}\Big|_a - \kappa_2 \frac{\partial\psi_2}{\partial r}\Big|_a \tag{3.6}$$

Let us consider separately the transport of charge at the interface, characterized by surface conductivity λ, with $\kappa = \kappa_2 = 0$. For an axially symmetric field distribution and a spherical surface, the continuity equation for process (b) is

$$\frac{d\sigma_b}{dt} = \frac{\lambda}{a^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi_2}{\partial\theta}\Big|_a\right) \tag{3.7}$$

Treating λ, κ and κ_2 as constants of the system, we consider that the time derivative of the surface charge density may be written as the sum of the two contributions (3.6) and (3.7), when processes (a) and (b) occur simultaneously. Thus

$$\frac{d\sigma}{dt} = \frac{\lambda}{a^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial\psi_2}{\partial\theta}\Big|_a\right) + \kappa\frac{\partial\psi}{\partial r}\Big|_a - \kappa_2\frac{\partial\psi_2}{\partial r}\Big|_a \tag{3.8}$$

This equation is solved readily by standard methods, giving these results:

The potential inside the sphere is

$$\psi_2 = -3\,\epsilon^* rE\cos\theta/(\epsilon_2^* + 2\epsilon^* - 8\pi i\lambda/\omega a) \tag{3.9}$$

The external potential is

$$\psi = [a^3(\epsilon_2^* - \epsilon^* - 8\pi i\lambda/\omega a)/r^3(\epsilon_2^* + 2\epsilon^* - 8\pi i\lambda/\omega a) - 1]rE\cos\theta \tag{3.10}$$

The density of free surface charge produced by the applied field is

$$\sigma = 3iE\cos\theta[\epsilon_2\kappa - \epsilon(\kappa_2 + 2\lambda/a)]/\omega[\epsilon_2^* + 2\epsilon^* - 8\pi i\lambda/\omega a] \tag{3.11}$$

where

$$\epsilon^* = \epsilon - 4\pi i\kappa/\omega \tag{3.12a}$$

$$\epsilon_2^* = \epsilon_2 - 4\pi i\kappa_2/\omega \tag{3.12b}$$

Examination of equations 3.9, 3.10 and 3.11 reveals that the term $8\pi i\lambda/\omega a$ is always subtracted from ϵ_2^*. Comparison with equation 3.12b shows that the effect of the surface conductivity λ in this model is exactly equivalent to an increase of κ_2 by $2\lambda/a$. Thus, for spheres, the equations are the same as in a treatment involving bulk conductivities only (Maxwell–Wagner polarization) if the conductivity of the sphere is given by

$$\kappa_s = \kappa_2 + \kappa_s' \tag{3.13}$$

where $\kappa_s' = 2\lambda/a$. Thus, κ_s may be regarded as the effective volume conductivity of a sphere electrically equivalent to our model. The important

(12) F. Urban, H. L. White and E. A. Strassner, THIS JOURNAL, **39**, 311 (1935).

(13) J. G. Kirkwood and J. B. Shumaker, *Proc. Natl. Acad. Sci.*, **38**, 855 (1952).

(14) C. J. F. Böttcher, *Rec. trav. chim.*, **64**, 47 (1945).

difference between the true volume and the surface contributions is that the latter κ_s' varies inversely as the radius, which permits distinguishing between surface and volume conductivities when the radius can be varied independently. The complex dielectric constant of the equivalent sphere is

$$\epsilon_s^* = \epsilon_2 - 4\pi i\kappa_s/a \tag{3.14}$$

From equation 3.9 it follows that the electric field within the sphere is

$$E_2 = \frac{3\epsilon^*}{\epsilon_s^* + 2\epsilon^*} E \tag{3.15}$$

The dipole moment m of the free surface charge is calculable from the equation

$$m = 2\pi a^2 \int_0^\pi a(\cos\theta)\sigma \sin\theta d\theta$$

By application of (3.11)

$$m = i4\pi a^3(\epsilon_2\kappa - \epsilon\kappa_s)E/(\epsilon_s^* + 2\epsilon^*) \tag{3.16}$$

4. Dielectric Increments and Conductivities of Suspensions of Spheres

Polder and van Santen[15] have given an expression for the real part of the dielectric constant, ϵ, of a suspension of randomly oriented ellipsoids of dielectric constant ϵ_2 in a medium of dielectric constant ϵ_1. For suspensions of spheres their equation becomes

$$\frac{\epsilon - \epsilon_1}{3\epsilon} = \delta_2 \frac{\epsilon_2 - \epsilon_1}{\epsilon_2 + 2\epsilon} \tag{4.1}$$

where δ_2 is the volume fraction of the spheres, which may be uniform or polydisperse. Although this equation was derived for insulating dielectrics where the ϵ are all real, it is valid also for lossy dielectrics or conducting media where the ϵ are complex. For the present problem the complex dielectric constants ϵ_s^* and ϵ_1^* may be inserted. Then the complex dielectric constant ϵ^* of the suspension is given by

$$\frac{\epsilon^* - \epsilon_1^*}{3\epsilon^*} = \delta_2 \frac{\epsilon_s^* - \epsilon_1^*}{\epsilon^* + 2\epsilon^*} \tag{4.2}$$

Other expressions based upon different treatments of the mixture problem may readily be obtained in a similar way.

From equations 3.13 and 3.14, it is evident that the complex dielectric constant of a sphere with surface conductivity depends upon λ. It also depends upon the size of the sphere unless λ is zero. Therefore equation 4.2 is applicable only to spheres of a given size and λ. If λ is constant or substantially independent of a, the complex dielectric constant of a heterodisperse suspension of spheres may be obtained by application of equation 13 of Polder and van Santen,[15] and equation 3.14 above, with the result

$$\epsilon^* = \epsilon_1^* \left[1 - \sum_i \delta_i(\epsilon_s^* - \epsilon_1^*)(\epsilon_s^* + 2\epsilon_1^*)/3\right]^{-1} \tag{4.3}$$

Here δ_i is the volume fraction of spheres of radius a, and the sum is taken over all sizes. For low concentrations, that is, $\Sigma\delta_i << 1$, this may be simplified to

$$\epsilon^* - \epsilon_1^* = (\epsilon_1^*/3)\Sigma\delta_i(\epsilon_s^* - \epsilon_1^*)(\epsilon_s^* + 2\epsilon_1^*) \tag{4.4}$$

(15) D. Polder and J. H. van Santen, *Physica*, **12**, 257 (1946).

For insulating media, *i.e.*, $\kappa_1 = 0$, it is readily confirmed that the relaxation time is given by $\tau = a(\epsilon_2 + 2\epsilon_1)/8\pi\lambda$, a result obtained earlier by a simpler method.[4] Schwan[16] has pointed out recently that this is equivalent to a result of Miles and Robertson.[10] These investigators treated the complex conductivity of a suspension of spheres with a uniformly conducting shell of finite thickness. In the limit of very thin shells, their equations for relaxation time and conductivity reduce to the above. This agreement confirms the validity of the differential equation 3.8 employed in the present treatment. Miles and Robertson did not discuss the amplitude of the dielectric dispersion, nor did they relate the parameters in their model to known ionic properties. Because of the variation of ionic concentrations within the ion atmosphere, the idealized surface layer employed here is probably as good a model as the thick uniform shell they treated, and it has the advantage of greater simplicity.

In very dilute suspensions, or $\delta_2 << 1$, equation 4.2 for spheres one size may be simplified by inserting ϵ_1^* for ϵ^* in the denominators on both sides, which yields an explicit equation for ϵ^*. Further simplifications occur in the limits of high and low frequencies. Thus, when $\delta_2 \to 0$, $\epsilon^* \to \epsilon_1^*$, and the following expressions can be obtained for the dielectric constant increment, $\Delta\epsilon = \epsilon - \epsilon_1$, and the dielectric loss increment $\Delta\epsilon'' = \epsilon'' - \epsilon_1'' = 4\pi i(\kappa - \kappa_1)/\omega$

$$\frac{\Delta\epsilon}{\delta_2} = 3\frac{\epsilon_1 X + 3\epsilon_1'' Y}{Z} \tag{4.5}$$

$$\frac{\Delta\epsilon''}{\delta_2} = 3\frac{\epsilon_1'' X - 3\epsilon_1 Y}{Z} \tag{4.6}$$

where

$$X = \epsilon_2^2 + \epsilon_1\epsilon_2 - 2\epsilon_1^2 + \epsilon_s''^2 + \epsilon_1''\epsilon_s'' - 2\epsilon_1''^2$$
$$Y = \epsilon_2\epsilon_1'' - \epsilon_1\epsilon_s''$$
$$Z = (\epsilon_2 + 2\epsilon_1)^2 + (\epsilon_s'' + 2\epsilon_1'')^2$$

In the limit of very low frequencies, where ionic conduction dominates the electric field distribution, it can be shown that

$$\left(\frac{\Delta\epsilon}{\delta_2}\right)_{\substack{\omega\to 0 \\ \delta_2\to 0}} = \frac{3[(r^2 - 2r - 2)\epsilon_1 + 3\epsilon_2]}{(r+2)^2} \tag{4.7}$$

where $r = \kappa_s/\kappa_1$. Thus, the low frequency dielectric increment is strongly dependent upon the conductivity ratio, which involves both the bulk conductivities and the surface conductivity. This is illustrated in Fig. 2, which is a plot of equation 4.7 for $\epsilon_1 = 78.54$ (H_2O, 25°) and $\epsilon_2 = 3.0$. Also, at low frequencies

$$\left(\frac{\Delta\kappa/\kappa_1}{\delta_2}\right)_{\substack{\omega\to 0 \\ \delta_2\to 0}} = 3\frac{r-1}{r+2} \tag{4.8}$$

The relative conductivity increment per unit volume fraction is $-3/2$ for $r = 0$, negative for $r > 1$, passes through zero for $r = 1$, and reaches the value 3.0 as $r \to \infty$, as shown by the dashed line of Fig. 2.

At frequencies well above the regions of dispersion, where the dielectric constants largely determine the field distribution

(16) H. P. Schwan, *Adv. Biol. and Med. Phys.*, **5**, 147 (1957).

$$\Delta\epsilon/\epsilon_1\delta_2 = 3(\epsilon_2^2 + \epsilon_1\epsilon_2 - 2\epsilon_1^2)/(\epsilon_2 + 2\epsilon_1)^2 \quad (4.9)$$
$$(\omega \to \infty,\ \delta_2 \to 0)$$

and

$$\Delta\kappa/\kappa_1\delta_2 = 3(\epsilon_2^2 - 2\epsilon_1\epsilon_2 - 2\epsilon_1^2 + 3\epsilon_1^2 r)/(\epsilon_2 + 2\epsilon_1^2) \quad (4.10)$$
$$(\omega \to \infty,\ \delta_2 \to 0)$$

From equation 4.9, it can be seen that the dielectric increment is independent of the conductivities where frequency is so high that the ionic currents are negligible compared to the displacement currents. In strong contrast, the conductivity increment increases with the conductivity ratio and may be very large even for a relatively low volume concentration. In this case, it may not be permissible to set $\epsilon^* = \epsilon_1^*$ in equation 4.2, except for very low concentrations. It is clear that measurements of conductivity increments at frequencies well above the dispersion region can yield important information regarding the mobilities and binding of counterions.

Employing equations 4.5 and 4.6, which are valid for any frequency, the dispersion curves were calculated assuming $\kappa_1 = 1.469 \times 10^{-4}$ (10^{-3} M KCl at 25°) and $\lambda/a = 2.0 \times 10^{-3}$ ohms^{-1} cm.$^{-1}$. The latter would correspond to 500 Å. radius spheres of $\lambda = 10^{-8}$ ohms^{-1}, or 100 Å. radius spheres of $\lambda = 2 \times 10^{-9}$ ohms^{-1}. Values of λ for glass surfaces in contact with water are of this order.[4,12] Assuming negatively charged spheres, with λ due mainly to K^+ counterions having their usual mobilities, the polyion valencies would be around 2500 or 200, respectively. The results are plotted in Fig. 3.

5. The Surface Contributions to the Effective Conductivities of Ellipsoids, Cylinders and Disks

In part 3, it was shown that the effect of a surface conductivity is the same as an increase of the volume conductivity by an amount $2\lambda/a$ for a sphere. We may anticipate that the surface contributions to the effective conductivities will be different along different directions in non-spherical particles. Because the electrical boundary value problems cannot be solved in closed form for irregular shapes, we shall confine the discussion to ellipsoids, for which closed solutions are available, and two figures of revolution, the cylinder and disk, which approximate limiting cases of the ellipsoid when the appropriate axial ratio is high. The latter models are of particular interest in connection with the electrical properties of solutions of large rigid rod-like macromolecules like the nucleic acids, charged α-helical polypeptides, tobacco mosaic virus, and of suspensions of plate-like colloidal particles like the bentonite clays.

To greatly simplify the calculations, we shall first see how the equation for the surface contribution, $\kappa_s' = 2\lambda/a$, to the effective conductivity of a sphere, can be obtained in an easy way, and how the procedure may be generalized to give approximate values of the surface contributions along the various axes of an ellipsoid.

Consider a section of the sphere of radius a which is perpendicular to the direction of the internal field, E_2. The radius of this section is $a \sin\theta$, where θ is the angle between a radius vector

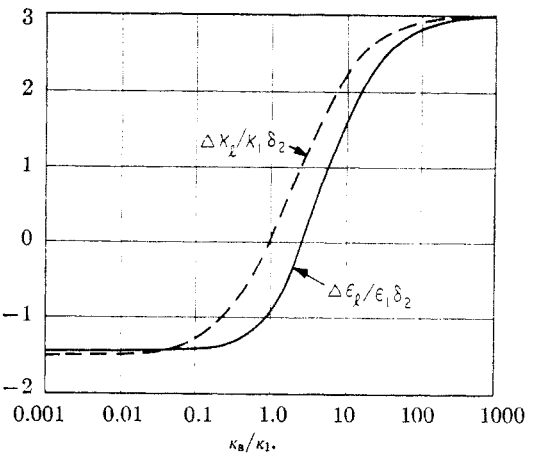


Fig. 2.—Low frequency dielectric and conductivity increments *vs.* conductivity ratio, κ_2/κ_1 for spheres.

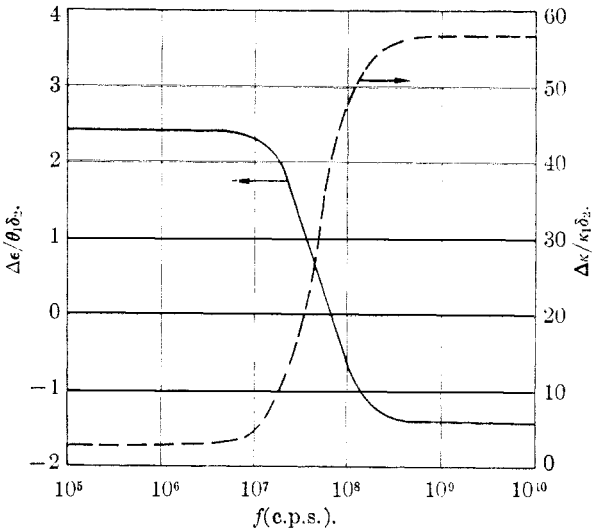


Fig. 3.—Dispersion of the dielectric constant and conductivity.

to the edge of the section and the field direction. The rate of charge transport along the surface across this section is $\lambda E_t 2\pi a \sin\theta$, where E_t is the tangential component of the internal field, equal to $E_2 \sin\theta$. The rate at which charge would be transported across the section as a result of a hypothetical volume conductivity κ_s' is $\kappa_s' E_2 A$, where A is the area of the section, equal to $\pi(a \sin\theta)^2$. Equating these two rates, one obtains $\kappa_s' = 2\lambda/a$, independent of the value of θ. Thus, the surface contribution to the effective volume conductivity is constant throughout the sphere, and may be evaluated readily by considering an equatorial section, *i.e.*, $\theta = \pi/2$.

Ellipsoids.—We next consider an ellipsoid with electric field along one of its semi-axes, a, b or c, to evaluate the surface contributions, κ_a', κ_b', κ_c', to the effective conductivity along the respective axes. Taking equatorial sections perpendicular to the respective axes, making the assumption that the field intensity is constant across the section, as it is known to be for a homogeneous ellipsoid, and equating the expressions for surface and volume charge transport, one readily obtains

$$\kappa_a' = C(b,c)\lambda/\pi bc \quad (5.1a)$$
$$\kappa_b' = C(a,c)\lambda/\pi ac \quad (5.1b)$$
$$\kappa_c' = C(a,b)\lambda/\pi ab \quad (5.1c)$$

Here $C(j,k)$ is the circumference of the ellipse with semi-axes j and k. This is approximately equal to $[2\pi(j^2 + k^2)/2]^{1/2}$ if it is nearly circular and approaches $4(j + k)$ if highly eccentric. These are equatorial values for κ'. Further consideration shows that the κ' values are not strictly constant, depending upon position along the axis, but that they are nearly constant for certain cases of special interest. Thus, for a slightly oblate or prolate spheroid, they approach the κ' evaluated for a sphere, shown to be a $2\lambda/a$ at all planes. For oblate or prolate ellipsoids of revolution ($b = c$)

$$\kappa_a' = 2\lambda/b \quad (5.2)$$

Thus, at the equator, the surface contribution to the effective conductivity along the symmetry axis equals that of a sphere of radius b. Further detailed consideration, in which the tangential component of the internal field was evaluated as a function of position along the surface, showed that $\kappa_a'(f) = (2\lambda/b)[1 - (1 - b^2/a^2)f^2]^{-1/2}$, where f is the fractional distance along the axis a from the center to the end of the ellipsoid. If $a>>b$, κ_a' goes from $2\lambda/b$ to 1.16 $(2\lambda/b)$ as the circular section goes from the equator to $f = 0.5$. This displacement encompasses most of the volume of the semi-ellipsoid; therefore, in the present state of our knowledge, it seems adequate to consider $\kappa_a'(f)$ a constant equal to $2\lambda/b$ for a prolate ellipsoid of revolution. If $a<<b$, the oblate ellipsoid approaches a disk, and $\kappa_a'(f) = (2\lambda/b)[1 + (b^2/af^2)^2]^{-1/2}$. In going from $f = 0$ to $f = 0.5$, $\kappa_a'(f)$ decreases from $2\lambda/b$ to $(a/b)(2\lambda/b)$. This is a greater change than the corresponding variation for the prolate ellipsoid, but it is not very important because, as we shall see, the transverse contribution, κ_b', is much greater than κ_a' and, in addition, positive contributions to the conductivity of the system will be much less for a circular disk with axis along the field than with axis perpendicular, as a result of the different depolarization factor A_j, defined below. It appears doubtful that a rigorous treatment, similar to the one above for a sphere, would lead to a closed solution of the boundary-value problem.

For prolate ellipsoids of revolution of high axial ratio ($a >>b = c$), the surface contributions to the effective conductivities along the transverse axes are, from equation 5.1

$$\kappa_b' = \kappa_c' \simeq 4\lambda/\pi b \quad (5.3)$$

For thin oblate ellipsoids, approaching disks ($a << b = c$)

$$\kappa_b' = \kappa_c' \simeq 4\lambda/\pi a \quad (5.4)$$

In both cases, the smallest dimension determines the equatorial value of κ_b'. Both values are greater than κ' for a sphere of equal volume and surface conductivity. For the long thin prolate ellipsoid of revolution, equations 5.2 and 5.3 yield $\kappa_a'/\kappa_b' \cong \pi/2$, whereas for thin discs we find $\kappa_a'/\kappa_b' \cong \pi a/2b$. Thus, the anistropy of the surface contribution to the effective conductivity is very weak for needles and very strong for disks. This is an interesting result from the point of view of relaxation behavior and particle shape studies. Because $\kappa_a' << \kappa_b'$ for a disk, use of the equatorial value for κ_b' is probably adequate for most purposes.

Cylindrical Symmetry: Rods and Disks.—Again considering the internal field to be homogeneous, *i.e.*, neglecting end effects, we may equate the surface rate of charge transport and the volume rates for a cylinder of length $2a$ and diameter $2b$. The resulting values for the axial and transverse surface contributions to the effective conductivity are

$$\kappa_a' = 2\lambda/b \quad (5.5)$$
$$\kappa_b' = (a + b)\lambda/ab \quad (5.6)$$

It is seen that $\kappa_a' = \kappa_b'$ when $a = b$. At first glance this appears of interest in steady or low-frequency studies when the solvent conductivity is adjusted to equal the effective conductivity of the particle, in which case the field is uniform and end effects might be presumed non-existent. But more detailed consideration reveals that the model is not justified for the case of $a \simeq b$ because of the fundamental spatial difference between the surface and volume transport processes, *i.e.*, end effects are probably important. In the degenerate cases, equations 5.5 and 5.6 should be good approximations, regardless of the ratio of particle and solvent conductivities. Equation 5.6 becomes

$$\text{Rods } (a>>b),\ \kappa_b' = \lambda/b \quad (5.7)$$
$$\text{Disks } (a<<b),\ \kappa_b' = \lambda/a \quad (5.8)$$

The ratio between these values of κ_b' and the corresponding ones from equations 5.3 and 5.4 is $4/\pi$. Although this is a small difference, there may be a preference, to be decided by the actual shape when it is known. For rods, $\kappa_a'/\kappa_b' = 2$, and for disks, $\kappa_a'/\kappa_b' = 2(a/b)$. The relatively small anisotropy for rods does not involve a and should be present even when the ion atmosphere is comparable in thickness to b, so long as it is thin compared to a.

6. Dielectric Constant and Conductivity Increments, Non-spherical Systems

Oriented Ellipsoids, Rods and Disks.—The equations of Fricke[11] for the conductivity of dilute suspensions of isotropic ellipsoids with zero surface conductivity may be modified to obtain the more general equations appropriate to the present problem. If the ellipsoids are oriented with axis $j = a$, b or c along the direction of the field, Fricke's equation 1 may be written in the form

$$\kappa - \kappa_1 = \delta_2(\kappa_j - \kappa_1)/[1 + A_j(\kappa_j - \kappa_1)/\kappa_1] \quad (6.1a)$$

Here, κ and κ_1 are the specific conductivities of the suspension and the solvent, respectively, κ_j is the particle conductivity along the j-axis, and A_j is the shape-dependent depolarization factor of the ellipsoid, given by

$$A_j = \int_0^\infty \frac{abc\ ds}{2(j^2 + s)[(a^2 + s)(b^2 + s)(c^2 + s)]^{1/2}} \quad (6.1b)$$

This is equal to $abcL_j/2$ in Fricke's notation, and seems more convenient and meaningful than his L_j or x_j. Furthermore, graphs[17] and special formulas[18] for this function are available.

(17) J. A. Osborn, *Phys. Rev.*, **67**, 351 (1945).

The complex dielectric constants may be substituted for the conductivities in equation 6.1, because the boundary-value problems involving the two are mathematically identical. Alternatively, the equations of Polder and van Santen[15] derived for the static dielectric constant can be shown to give the same result in the limit of very low concentrations, after inserting the complex dielectric constants. Thus, one obtains for the dilute suspension, oriented with j-axis along the field

$$\epsilon^* - \epsilon_1^* = \delta_2(\epsilon_j^* - \epsilon_1^*)/[1 + A_j(\epsilon_j^* - \epsilon_1^*)/\epsilon_1^*] \quad (6.2)$$

Here, each complex dielectric constant, ϵ_k^*, may be written

$$\epsilon_k^* = \epsilon_k - 4\pi i\kappa_k/\omega, \; k = 1, a, b, \text{ or absent} \quad (6.3)$$

If the media exhibit dispersions, ϵ_k is the real value of the dielectric constant at the particular frequency, and κ_k would include the loss from the dispersive process. For the ellipsoid

$$\epsilon_j^* = \epsilon_j - 4\pi i\kappa_j/\omega \quad (6.4)$$

$$\kappa_j = \kappa_j^0 + \kappa_j' \quad (j = a, b \text{ or } c) \quad (6.5)$$

where κ_j^0 is the true volume conductivity along the j-axis, and κ_j' is the surface contribution to κ_j, the effective conductivity. For ellipsoids, values of κ_j' may be obtained from equation 5.1 or 5.2 to 5.4. If the particles are cylindrical rods or flat disks, equations 5.6 to 5.8 may be employed for the surface contributions, and the value of A_j may be taken as that of an ellipsoid of revolution of the same axial ratio. The volume fraction should be the actual external volume of the particles, determined experimentally.

Random Orientation.—For a dilute suspension of randomly oriented ellipsoids, equation 6.2 becomes

$$\epsilon^* - \epsilon_1^* = \frac{\delta_2}{3}\sum_{j=a,b,c}\frac{\epsilon_j^* - \epsilon_1^*}{1 + A_j(\epsilon_j^* - \epsilon_1^*)/\epsilon_1^*} \quad (6.6)$$

Equations applicable to more concentrated ellipsoidal systems are obtained easily by starting with Polder and van Santen's[15] equation 13. They are not presented as they cannot be solved explicitly for the dielectric constant and conductivity. Their usefulness would be doubtful because (a), the number of variables in ellipsoidal systems becomes large, and (b), ion atmosphere properties will become concentration dependent in an unknown manner when the ion atmospheres begin to overlap. Experimental tests of this theory should be confined to dilute systems, preferably by extrapolation of specific dielectric and conductivity increments to zero concentration.

The real and imaginary terms of equation 6.6 may be separated, after appropriate manipulation, to obtain the dielectric constant increment

$$\Delta\epsilon/\epsilon_1 = (\epsilon - \epsilon_1)/\epsilon_1 \quad (6.7)$$

and the dielectric loss increment

$$\Delta\epsilon''/\epsilon_1'' = (\epsilon'' - \epsilon_1'')/\epsilon_1'' = (\kappa - \kappa_1)/\kappa_1 = \Delta\kappa/\kappa_1 \quad (6.8)$$

The resulting equations are

$$\Delta\epsilon/\epsilon_1 = (\delta_2/3)\sum_{j=a,b,c}(\epsilon_j/\epsilon_1 - 1 + N_j)/D_j \quad (6.9)$$

and

$$\Delta\epsilon''/\epsilon_1'' = (\delta_2/3)\sum_{j=a,b,c}(\epsilon_j''/\epsilon_1'' - 1 + N_j)/D_j \quad (6.10)$$

where

$$N_j = [(\epsilon_j - \epsilon_1)^2 + (\epsilon_j'' - \epsilon_1'')^2]A_j/(\epsilon_1^2 + \epsilon_1''^2)$$

and

$$D_j = [1 + \{(\epsilon_j\epsilon_1 + \epsilon_j''\epsilon_1'')/(\epsilon_1^2 + \epsilon_1''^2) - 1\}A_j]^2 + [(\epsilon_j''\epsilon_1 - \epsilon_j\epsilon_1'')A_j/(\epsilon_1^2 + \epsilon_1''^2)]^2$$

Numerical calculations of the dispersion may be made from these equations after appropriate simplification for a given system. Fricke's equations are for $\epsilon_a = \epsilon_b = \epsilon_c = \epsilon_2$, and $\epsilon_a'' = \epsilon_b'' = \epsilon_c'' = \epsilon_2''$, that is, he did not consider anisotropic particles. As in his case there are three relaxation times for a given ellipsoid, corresponding to the three axes. By generalization of Fricke's equation[11] 14 for anisotropic particles, the relaxation times in dilute systems are

$$\tau_j = \frac{1}{4\pi}\frac{\epsilon_j + \epsilon_1(1/A_j - 1)}{\kappa_j + \kappa_1(1/A_j - 1)} \quad (6.11)$$

where κ_j, in e.s.u., is given by equation 6.5. In the case of Maxwell–Wagner polarization, that is, zero surface conductivity, these are dependent on the shape of the particle, but independent of size. In our model they depend both upon size and shape, since κ_j', the surface contribution to the effective conductivity, clearly depends upon size as well as shape. Thus, randomly oriented polydisperse systems may be expected to show three broadened regions of dielectric dispersion, or two if there is an axis of symmetry.

If the frequency is low enough so that all dielectric constants are small relative to the corresponding loss factors, the fields are determined by the conductivities, and the equation for the dielectric increment, $\Delta\epsilon_l$, reduces to

$$\frac{\Delta\epsilon_l}{\epsilon_1} = \frac{\delta_2}{3}\sum_{j=a,b,c}\frac{\epsilon_j/\epsilon_1 - 1 + (\kappa_j/\kappa_1 - 1)^2A_j}{[1 + (\kappa_j/\kappa_1 - 1)A_j]^2} \quad (6.12)$$

The low frequency conductivity increment is

$$\frac{\Delta\kappa_l}{\kappa_1} = \frac{\delta_2}{3}\sum_{j=a,b,c}\frac{\kappa_j/\kappa_1 - 1}{1 + (\kappa_j/\kappa_1 - 1)A_j} \quad (6.13)$$

At sufficiently high frequencies so that loss factors are all small relative to the corresponding dielectric constants

$$\frac{\Delta\epsilon_h}{\epsilon_1} = \frac{\delta_2}{3}\sum_{j=a,b,c}\frac{\epsilon_j/\epsilon_1 - 1}{1 + (\epsilon_j/\epsilon_1 - 1)A_j} \quad (6.14)$$

and

$$\frac{\Delta\epsilon_h''}{\epsilon_1} = \frac{\Delta\kappa_h}{\kappa_1} = \sum_{j=a,b,c}\frac{\kappa_j/\kappa_1 - 1 + (\epsilon_j/\epsilon_1 - 1)^2A_j}{[1 + (\epsilon_j/\epsilon_1 - 1)A_j]^2} \quad (6.15)$$

For the present model and the Maxwell–Wagner model, no allowance is made for the possibility of the applied field causing large modifications of ion distributions at the boundaries, so the equations are strictly valid only for small fields. Fricke has further restricted application of his high frequency equations to frequencies "above all dispersion regions." If this is interpreted literally, his equations 22 and 23 and our 6.14 and 6.15 could only be applied above the dipolar dispersion region of polar solvents which would seriously restrict their use, for example, in aqueous systems. In obtain-

(18) H. C. van de Hulst, "Light Scattering by Small Particles," John Wiley and Sons, Inc., New York, N. Y., 1957.

ing equations 6.14 and 6.15 from 6.9 and 6.10 here, it was seen that this restriction is not necessary. If one stays below the dipolar dispersion region of the solvent, where the dipolar contribution to ϵ_1'' is always much less than ϵ_1, the equations for $\Delta\epsilon_h$ and $\Delta\epsilon_h''$ are applicable. Thus, they may be applied for dilute aqueous solutions up to around a few hundred megacycles. When the conductivities are so high that ion atmosphere and dipolar dispersion regions overlap significantly, the more exact expressions 6.9 and 6.10 should be used. It should be kept in mind that the equations are for dilute systems, *i.e.*, $\Delta\epsilon < \epsilon_1$ and $\Delta\kappa < \kappa_1$.

Numerical Results.—To illustrate the various effects of size, shape, magnitude of the surface conductivity, and solvent conductivity upon the dielectric and conductivity increments, numerical calculations were made for a wide range of values of the important parameters of the system. These are directed primarily at elucidation of the properties of rigid particles of relatively low dielectric constant and bulk conductivity, in a medium of high dielectric constant and low to moderate conductivity.

First, to illustrate the effect of particle shape on the primary quantities appearing in the equations, values of the depolarization factors A_a and $A_b = A_c$, and of the surface contributions to the effective conductivity, κ_a' and $\kappa_b' = \kappa_c'$, are presented in Table I. These apply to ellipsoids of revolution, where $2a$ is the symmetry axis and $2b = 2c$, with axial ratio $p = a/b$. The surface conductivity λ and the particle volume were held constant, and equations 5.1a to -c were used. The κ' values are given in units of κ_s', the value for a sphere of the same volume and surface conductivity. It is seen that κ_a' increases monotonically with increasing axial ratio, whereas κ_b' goes through a minimum at p slightly greater than one, increasing markedly for axial ratios much less than one, and somewhat less for axial ratios much greater than one. The important depolarizing factors are the smaller ones, for it is along the corresponding axis that the field exerts its greater effect.

TABLE I

DEPOLARIZING FACTORS AND SURFACE CONTRIBUTIONS TO THE EFFECTIVE CONDUCTIVITY *vs.* AXIAL RATIO OF SPHEROIDS

$p = a/b$	A_a	A_b	κ_a'/κ_s'	κ_b'/κ_s'
200	0.0001333	0.4999	5.848	3.724
100	.0004666	.4997	4.642	2.956
50	.001466	.4992	3.684	2.348
30	.003466	.4982	3.107	1.983
20	.006733	.4966	2.714	1.736
10	.02026	.4898	2.154	1.393
5	.05579	.4720	1.710	1.144
3	.1085	.4457	1.442	1.022
3/2	.2330	.3834	1.145	0.963
1	.3333	.3333	1.000	1.000
2/3	.4459	.2770	0.8735	1.103
1/3	.6354	.1822	.6933	1.475
1/10	.8607	.06959	.4641	3.002
1/30	.9497	.02509	.3218	6.161
1/100	.9844	.007766	.2154	13.72
1/200	.9922	.003910	.1710	21.77

Low frequency conductivity increments of randomly oriented systems were computed from equation 6.13 for a system of particles with zero volume conductivity, for varying axial ratio, again keeping the particle volume $(4\pi a_0^3/3)$ constant, for various values of $2\lambda/a_0\kappa_1 = \kappa_s'/\kappa_1$. Results are presented in Table II. No assumption regarding dielectric constants was required. For $\kappa_s'/\kappa_1 = 1$, the field is not affected by insertion of the sphere for its effective conductivity is equal to that of the medium, and $\Delta\kappa = 0$. But with highly elongated particles, the combination of small depolarization factor and increased effective conductivity along the longer axis produces positive conductivity increments. These increase rapidly for large values of p or $1/p$, and, of course, increase with κ_s'/κ_1. Thus, very large contributions to the low-frequency conductivity may occur in relatively dilute systems of highly anisometric particles with surface conductivity. The numerical calculation in Part 4 for spheres with $\lambda/a_0 = 2 \times 10^{-3}$ ohms^{-1} cm.$^{-1}$ in 10^{-3} M KCl corresponds to $\kappa_s'/\kappa_1 = 27$.

For oriented systems, the conductivity increments depend upon whether the field is oriented along the a or the b-axis, and these are designated $\Delta\kappa_a$ and $\Delta\kappa_b$, respectively. Values for parallel orientation are given in Table III. They are obtained by taking the appropriate term of the sum of equation 6.13, with the coefficient δ_2 rather than $\delta_2/3$. It is apparent that an anisotropy of the conductivity of the solution will be produced by partial orientation of the particles, and its magnitude can be obtained from these values if the degree of orientation is known.

High frequency conductivity increments for a broad range of p and κ_s'/κ_1 are given in Table IV. Equation 6.15 was employed, with $\epsilon_1 = 78.54$ (H_2O, 25°) and $\epsilon_a = \epsilon_b = \epsilon_c = 3.00$. The increments are only weakly dependent upon the latter quantity, for small ϵ_j/ϵ_1. Comparison with Table II shows that $\Delta\kappa_h$ is greater than $\Delta\kappa_l$ for large κ_s'/κ_1. This is because the local field remains large at high frequencies, but approaches zero at low frequencies for high κ_s'/κ_1. In oriented systems, $\Delta\kappa_a$ and $\Delta\kappa_b$ were computed from the appropriate term of the sum in equation 6.15. They are presented in Table V. Again, large anisotropies are observed.

The high frequency dielectric increment is independent of conductivity ratio, as seen by equation 6.14. Table VI lists values of $\Delta\epsilon_h/\epsilon_1\delta_2$ calculated for $\epsilon_1 = 78.54$ and $\epsilon_a = \epsilon_b = \epsilon_c = 3.00$. It is negative and becomes more negative at very low axial ratios, as expected from considerations of the dielectric constants of insulating particles.[15] The dielectric increments of oriented systems also are listed in Table VI.

The low frequency dielectric increments were computed for the usual axial ratios, with κ_s'/κ_1 values ranging from 1 to 100, $\epsilon_1 = 78.54$, $\epsilon_j = 3.00$, $\kappa_j^0 = 0$ and are shown in Table VII. Additional p-values were computed to delineate the interesting maximum which was found at the higher κ_s'/κ_1 values and high p. At $\kappa_s'/\kappa_1 = 1$, the low frequency dielectric increment is always negative in the range $^1/_{100} \leqslant p \leqslant 100$. Insertion of

TABLE II

LOW FREQUENCY CONDUCTIVITY INCREMENTS *vs.* AXIAL RATIO FOR SPHEROIDS[a]

p	κ'_s/κ_1 = 0.1	0.3	1.0	3.0	10	30	100
100	−0.9032	+0.05089	1.871	5.345	16.07	44.59	128.3
30	−1.120	− .3610	1.137	3.648	10.29	24.59	51.11
10	−1.258	− .6614	0.5959	2.485	6.656	10.56	14.71
3	−1.312	− .8703	.1555	1.532	3.025	3.913	4.350
3/2	−1.294	− .9100	.02190	1.248	2.350	2.858	3.071
1	−1.286	− .9130	.0000	1.200	2.250	2.719	2.912
2/3	−1.300	− .9142	.02195	1.252	2.350	2.850	3.058
1/3	−1.465	− .9452	.1643	1.619	3.030	3.733	4.039
1/10	−2.264	−1.174	.8400	3.525	6.702	8.604	9.518
1/30	−4.247	−1.567	2.411	8.089	16.27	22.15	25.29
1/100	−8.526	−1.907	6.570	20.23	44.32	65.64	78.80

[a] The numbers are relative values per unit volume fraction, $\Delta\kappa_1/\kappa_1\delta_2$.

TABLE III

LOW FREQUENCY CONDUCTIVITY INCREMENTS OF ORIENTED SPHEROIDS

p	κ'_s/κ_1 = 0.1 a[a]	0.1 b[a]	1.0 a	1.0 b	3.0 a	3.0 b	10 a	10 b	100 a	100 b
100	− 0.536	−1.09	3.64	0.989	12.8	1.60	44.5	1.87	381	1.99
30	− .691	−1.33	2.09	.660	8.09	1.43	27.2	1.81	149	1.99
10	− .797	−1.49	1.13	.330	4.92	1.27	14.5	1.76	40.1	2.01
3	− .943	−1.50	0.422	.0223	2.44	1.08	5.46	1.80	8.66	2.20
3/2	− 1.12	−1.38	.140	− .0371	1.55	1.10	3.04	2.00	4.14	2.54
1	− 1.29	−1.29	.000	.000	1.20	1.20	2.25	2.25	2.91	2.91
2/3	− 1.54	−1.18	− .134	.100	0.941	1.41	1.74	2.66	2.19	3.49
1/3	− 2.28	−1.06	− .381	.437	.640	2.11	1.24	3.92	1.54	5.29
1/10	− 5.32	−0.736	− .995	1.76	.293	5.14	0.881	9.61	1.13	13.7
1/30	−12.0	− .388	−1.91	4.57	− .0358	12.2	.714	24.1	1.02	37.4
1/100	−26.6	+ .371	−3.45	11.6	− .543	30.6	.540	66.2	0.968	118

[a] Column a contains values of $\Delta\kappa_a/\kappa_1\delta_2$, where $\Delta\kappa_a$ is the conductivity increment for orientation of the a-axis along the field; column b contains $\Delta\kappa_b/\kappa_1\delta_2$ for orientation of the b-axis along the applied field.

TABLE IV

HIGH FREQUENCY CONDUCTIVITY INCREMENTS, $\Delta\kappa_h/\kappa_1\delta_2$, *vs.* AXIAL RATIO

p	κ'_s/κ_1 = 0.1	1.0	3.0	10	100
100	--0.7770	7.192	24.90	86.88	883.7
30	--1.068	4.256	16.09	57.49	589.9
10	--1.237	2.424	10.56	39.03	405.1
3	--1.305	1.113	6.488	25.30	267.2
3/2	--1.289	0.7316	5.222	20.94	223.0
1	--1.282	.6679	5.000	20.16	215.1
2/3	--1.296	.7370	5.254	21.06	224.3
1/3	--1.426	1.250	7.197	28.01	295.6
1/10	--2.256	4.509	19.54	72.16	748.7
1/30	--4.223	12.54	49.78	180.1	1856
1/100	--7.740	23.45	92.73	335.2	3453

spheres of the same effective conductivity as the solvent does not affect the field distribution, but the dielectric constant decreases because of the lower dielectric constant of the sphere. The effect of a lower internal dielectric constant is overcome at sufficiently high values of κ_a' and κ_b', so the increment becomes positive at higher values of κ_s'/κ_1, the more so when $p \neq 1$. Large positive values at low and high p are associated with low values of the depolarizing factor, and arise from the terms containing κ_j in equation 6.12. This is of particular interest regarding the interpretation of the large dielectric increments of anisometric polyelectrolytes. If κ_j/κ_1 is large enough so $\kappa_j/\kappa_1 >> 1$, the corresponding term of the sum approaches

TABLE V

HIGH FREQUENCY CONDUCTIVITY INCREMENTS OF ORIENTED SPHEROIDS[a]

p	κ'_s/κ_1 = 0.1 a	0.1 b	1.0 a	1.0 b	3.0 a	3.0 b	10 a	10 b	100 a	100 b
100	− 0.536	−0.898	3.65	8.97	12.9	30.9	45.5	108	464	1094
30	− .691	−1.26	2.12	5.32	8.38	19.9	30.3	71.1	312	729
10	− .797	−1.46	1.22	3.03	5.70	13.0	21.4	47.9	223	496
3	− .942	−1.49	0.676	1.33	4.27	7.60	16.9	29.5	179	311
3/2	− 1.11	−1.38	.598	0.798	4.40	5.63	17.7	22.6	189	240
1	− 1.28	−1.28	.668	.668	5.00	5.00	20.2	20.2	215	215
2/3	− 1.53	−1.18	.877	.667	6.23	4.77	25.0	19.1	266	204
1/3	− 2.27	−1.01	1.86	.946	11.0	5.28	43.1	20.5	456	216
1/10	− 5.31	−0.730	8.78	2.37	40.1	9.27	150	33.4	1559	344
1/30	−11.9	− .379	26.7	5.44	113	18.4	413	63.7	4277	646
1/100	−24.0	+ .370	44.5	12.9	197	40.8	729	138	7576	1392

[a] The numbers in column a are $\Delta\kappa_a/\kappa_1\delta_2$, in column b, $\Delta\kappa_b/\kappa_1\delta_2$.

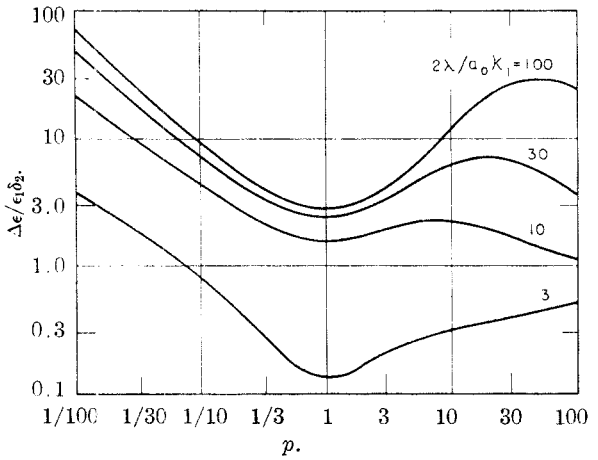


Fig. 4.—Dielectric increments *vs.* axial ratio, p, for ellipsoids of revolution. Particle volume conductivity is zero. The surface conductivity and the particle volume are held constant as p changes.

TABLE VI

HIGH FREQUENCY DIELECTRIC DECREMENTS FOR COMPLETE AND RANDOM ORIENTATION

p	$-\Delta\epsilon_a/\epsilon_1\delta_2$	$-\Delta\epsilon_b/\epsilon_1\delta_2$	$-\Delta\epsilon_h/\epsilon_1\delta_2$
100	0.9622	1.852	1.555
30	.9650	1.847	1.553
10	.9809	1.818	1.539
3	1.074	1.683	1.480
3/2	1.240	1.524	1.429
1	1.416	1.416	1.416
2/3	1.684	1.311	1.435
1/3	2.473	1.166	1.602
1/10	5.586	1.031	2.549
1/30	11.11	0.9856	4.360
1/100	18.08	0.9690	6.671

TABLE VII

LOW FREQUENCY DIELECTRIC INCREMENT, $\Delta\epsilon_1/\epsilon_1\delta_2$, *vs.* AXIAL RATIO

	κ_S'/κ_1				
p	1.0	3.0	10	30	100
200	−0.009257	0.7073	1.025	2.307	14.08
100	− .1554	.5305	1.162	3.617	23.66
50	− .3277	.4502	1.399	5.374	29.04
30	− .4553	.3965	1.681	6.708	27.01
20	− .5543	.3571	1.934	7.217	21.90
10	− .7128	.3109	2.265	6.390	12.18
5	− .8445	.2620	2.206	4.472	6.247
3	− .9139	.2133	1.970	3.329	4.142
3/2	− .9577	.1482	1.683	2.568	2.975
1	− .9618	.1337	1.628	2.454	2.826
2/3	− .9618	.1510	1.690	2.566	2.965
1/3	− .9837	.2710	2.131	3.322	3.900
1/10	−1.427	.8075	4.420	7.415	9.088
1/30	−1.534	1.818	9.702	18.20	23.76
1/100	−2.125	3.821	22.56	49.95	72.05
1/200	−4.473	6.365	35.63	88.25	136.8

$1/A_j$, which is a large quantity along the long axes of highly anisometric particles. The maximum in $\Delta\epsilon_1$ observed for $p > 1$ is augmented and shifted to larger p at greater κ_j/κ_1. Thus, of conductivity associated with the counterions of a polyelectrolyte is a very important factor in determining $\Delta\epsilon_1$—one often ignored in past discussions with reference to theories developed for insulating systems.

The low frequency dielectric increments, $\Delta\epsilon_a$ and $\Delta\epsilon_b$, of oriented systems are presented in Table VIII. Again, $\epsilon_1 = 78.54$, $\epsilon_j = 3.00$, and $\kappa_j^0 = 0$ for all axes.

The numerical calculations have been presented in the form of tables as the wide range of values encountered would require an excessive number of curves for interpolations of reasonable accuracy. However, because low frequency dielectric increments have been widely investigated, curves of $\Delta\epsilon_1/\epsilon_1\delta_2$ *vs.* p for various κ_s'/κ_1 are shown in Fig. 4 to portray the general trends and to facilitate interpolations over the wide range of values.

7. Applications

Proteins, Nucleic Acids and Nucleoproteins.—Dielectric dispersion has been studied in many protein solutions. The earlier work, which has been reviewed in several places,[19,20] was interpreted in terms of equations developed for insulating systems. The possibility of ionic polarization mechanisms, and variables like solvent conductivity, macromolecular charge and configuration, and counterion composition, which effect their contributions, were ignored, and the dielectric increments were attributed solely to permanent dipole orientation effects. Dielectric relaxation times were interpreted in terms of the rotational diffusion constants of rigid hydrodynamic models. From the above equations and numerical calculations, it is clear that the ion atmosphere polarization may account for the same effects previously attributed to orientation polarization. Thus, the existence of two relaxation times can be explained on the basis of a spheroidal model with surface conductivity as an alternative to one carrying a non-axial dipole moment. Either idea appears equally acceptable qualitatively and, in general, the possibility of contributions from both must be considered. However, there are several instances where the permanent dipole orientation theory is clearly inapplicable, namely, tobacco mosaic virus, hemocyanin and desoxyribonucleate.

An early suggestion of this came from a study of the Kerr effect in solutions of a large rigid macromolecule, tobacco mosaic virus (TMV), which showed very strong electric orientation which persisted at frequencies far above the expected orientational dispersion region and which was dependent upon electrolyte concentration.[21] The role of the ion atmosphere was qualitatively suggested on the basis of that study. The observed influence of electrolyte concentration can be explained through its effect on the parameters κ_j/κ_1 which strongly affect the dielectric increments, as shown above. A more quantitative treatment of the orientation effects on the present model involves additional considerations which will be discussed separately.[22] A discussion of the limiting case when κ_j/κ_1 is very large (κ_1 negligible) is con-

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TABLE VIII

LOW FREQUENCY DIELECTRIC INCREMENTS FOR ORIENTED SPHEROIDS[a]

	κ_s'/κ_1 = 1.0		3.0		10		30		100	
p	a	b	a	b	a	b	a	b	a	b
200	−0.957	+0.492	−0.921	1.52	−0.514	1.79	2.99	1.96	38.3	1.97
100	−.952	+.243	−.873	1.23	−.000655	1.74	7.02	1.91	67.0	1.97
50	−.944	−.0196	−.790	1.07	+.833	1.68	12.3	1.89	83.2	1.97
30	−.933	−.217	−.682	0.936	+1.78	1.63	16.4	1.88	77.1	1.97
20	−.921	−.371	−.563	.817	+2.63	1.59	17.9	1.86	61.8	1.97
10	−.893	−.623	−.290	.610	+3.78	1.50	15.5	1.85	32.6	1.98
5	−.864	−.835	−.00674	.396	+3.75	1.44	9.68	1.87	14.7	2.04
3	−.856	−.943	+.129	.256	+3.08	1.42	6.12	1.93	8.13	2.15
3/2	−.896	−.989	+.170	.137	+2.07	1.49	3.36	2.17	3.98	2.47
1	−.962	−.962	+.134	.134	+1.63	1.63	2.46	2.46	2.83	2.83
2/3	−1.07	−.907	+.0705	.191	+1.30	1.89	1.89	2.91	2.13	3.38
1/3	−1.39	−.780	−.0777	.445	+0.941	2.73	1.35	4.31	1.50	5.10
1/10	−3.23	−.526	−.464	1.44	+.611	6.32	0.972	10.6	1.10	13.1
1/30	−4.14	−.230	−1.03	3.24	+.384	14.4	.826	26.9	0.984	35.2
1/100	−6.86	+.244	−1.97	6.72	+.0765	33.8	.705	74.6	.920	108
1/200	−14.7	+.620	−0.330	9.71	−.159	144	.159	132	.889	205

[a] Columns a and b refer to $\Delta\kappa_a/\kappa_1\delta_2$ and $\Delta\kappa_b/\kappa_1\delta_2$, for orientation of the axes a and b, respectively, along the applied field.

tained in the more recent and complete study of the Kerr effect in TMV solutions.[23]

The dielectric properties of aqueous solutions of hemocyanin and sodium hyaluronate have been studied by Jacobson and co-workers.[24-26] They noted that in these systems also, the dielectric relaxation times were too short to be explained reasonably on the basis of dipolar reorientation. In studies of sodium thymonucleate partially oriented by hydrodynamic shear,[24] it was observed that the low frequency specific dielectric increments were very high. They increased with increasing velocity gradient when the electric field was along the stream lines, and decreased when the field was perpendicular. The large dielectric increments are expected on the basis of the present model, as are the changes in dielectric increment of the partially oriented systems. See Tables VII and VIII for large p and κ_s'/κ_1. Since the importance of the excess conductivity associated with the ion atmosphere and of solvent conductivity was not recognized, and the preparations studied were not well characterized, the necessary data for a quantitative interpretation on the present model are not available. Qualitative interpretations of this work subsequently were given by Jacobson,[27,28] who proposed that polyelectrolytes orient water molecules in extended ice-like hydration shells. We shall refer to this as the iceberg model. The oriented water was presumed to have a high value of the low frequency dielectric constant and dispersion properties similar to ice. It already has been pointed out on quantitative grounds that this model is inadequate to explain the electric orientation of TMV as an unreasonably thick shell of ice would be required.[23] Further tests of this proposal can be made by application of the equations above, taking ionic conductivities to be negligible. Thus, equation 6.14 may be used to compute the low frequency dielectric increment. If one allows for the possibility of the existence of moderately thick shells, a reasonable upper value for the effective dielectric constant of the macromolecule with iceberg would be that of ice at 0°, which is 91.5.[29] The shape of Helix Pomatia hemocyanin approximates that of a prolate spheroid with axial ratio about 10.[25,30] From equation 6.14, taking ϵ_j(eff.) = 91.5, ϵ_1 = 78.54, and p = 10, one obtains $\Delta\epsilon/\epsilon_1\delta_2$ = 0.16. The reported[25] dielectric increment per gram of hemocyanin per 100 ml. of solution was 3.7 at 20°. Taking a partial specific volume of 0.7 cc./g. for the protein, $\Delta\epsilon/\epsilon_1\delta_2 = 3.7/(80.3)(7 \times 10^{-3}) = 6.6$. Thus, the iceberg model would require that around 6.6/0.16 − 1 or about 40 cc. of water be converted to ice per cc. of protein, which corresponds to a shell hundreds of ångströms thick. Reduction of this to a reasonable solvation value could be achieved only with unreasonably high assumed values for the dielectric constant of the ice.

From thermodynamic, viscosity and dielectric relaxation studies of aqueous solutions of simple electrolytes, it is well established that the effect of the high electric fields prevailing near ions is to disrupt the strongly hydrogen-bonded water structure[31] and decrease the local viscosity.[32-35] There appears to be no sound basis for supposing that a reversal of this effect, involving large masses of water, would occur in the vicinity of polyelectrolytes. Furthermore, ions are known to cause a lowering of the low-frequency dielectric constants

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of solvated water.[34,35] The current view is that the first solvation shell of a small ion is radially oriented, the second is a disrupted, partially hydrogen-bonded structure, and beyond that a relatively normal water structure prevails.[35] Thus, not only is the iceberg hypothesis unnecessary but it is in contradiction with other observations as well.

Klotz[36] recently has endorsed Jacobson's iceberg hypothesis, apparently without any quantitative test, and has attempted to correlate many other observations on the basis of this model. The above considerations appear to remove the original justification for the hypothesis, and raise serious doubts regarding the validity of any interpretations of other phenomena in terms of thick ice-like shells. The long-range solvation hypothesis was actually proposed in somewhat different forms by Ostwald,[37] Hauser,[38] and Freundlich,[39] to explain the formation and the rheological properties of gels about three decades ago, only to be rejected later after more careful consideration.[40,41] The results of a recent X-ray study of hyaluronic acid by Laurent[42] constitute further evidence against the iceberg model.

To explain the larger dielectric increments of the more anisometric macromolecules, Jacobson required the additional special assumption that they crystallized out a greater amount of water than do the globular proteins. No special hypothesis is required to explain this in terms of the present model, as can be seen by reference to Fig. 4, where high dielectric increments are observed at high κ_s'/κ_1, for values of p far removed from unity.

A dielectric study of sodium thymonucleate by Jungner and Jungner[43] revealed large dielectric increments, and relaxation times shorter than expected on the basis of reorientation of the long axis of the macromolecule. These were interpreted as indicating zero dipole moment along the long axis and a transverse component which was very large, about $2 \times 10^4\ d$. This appeared consistent[44] with the magnitude of the intrinsic Kerr constant reported by Benoit.[45] But it is not consistent with the negative sign of the birefringence.[46] The macromolecule has a negative optical anisotropy factor,[47] so the direction of orientation is with the long axis along the electric field, contrary to the orientation expected for a large transverse dipole moment. Since the macromolecule is known to be highly charged, it seems clear that all of the observed electrical properties can be explained in terms of ion atmosphere polarization. This does not rule out the existence of a transverse dipole moment, but sets an upper limit to its value. To determine its magnitude would require further studies through which the effects of induced and permanent moment contributions would be separable. There now appears to be some possibility of this, by the application of the new birefringence saturation[48] and reversing field[23,49] techniques.

Proteins of smaller size than hemocyanin may have orientational relaxation times in the same frequency range as the ionic relaxation times. Then it becomes difficult to separate the contributions from the two processes from dielectric studies alone. If a more general theory combining both polarization mechanisms becomes available, it might become feasible to do this. In any event, studies as a function of pH and electrolyte composition should prove helpful. In deionized systems, the charge carriers will be H^+ and OH^-, and solvent conductivity will be relatively low. Thus, even if the acid and base groups are weakly dissociated, κ_s'/κ_1 may still be large. But the relaxation times increase with decreasing κ_s', and the ion atmosphere becomes more diffuse in salt-free solutions. Under certain conditions, the ion atmosphere relaxation time might be large compared to the reorientational relaxation times; then displacement currents will predominate over conduction currents in the reorientational dispersion region. When this is true, the previous interpretations of dielectric relaxation times[19,20] may become approximately correct. Application of theories developed for insulators may be possible for low electrolyte solutions of small spherical proteins, but each system should be subjected to appropriate tests before such theories can be applied with any degree of confidence.

A criterion of the applicability to aqueous solutions of theories developed for insulating systems is proposed. It is that $\tau_i > \tau_d$, where τ_i is the ionic relaxation time and τ_d is the orientational relaxation time. From the Debye–Stokes relation, one obtains, for spheres in water at 25°

$$\tau_d = 2.72 \times 10^{12}\ a_0^3 \qquad (7.1)$$

Applying equation 6.11 to a sphere, with the assumption that κ_1 is low, taking $\epsilon_1 = 78.5$ and $\epsilon_j = 3$

$$\tau_i = 7.1 \times 10^{-12}\ a_0/\lambda\ (\lambda \text{ in ohms}^{-1}) \qquad (7.2)$$

Thus, $\tau_d/\tau_i = 3.8 \times 10^{23}\ a_0^2/\lambda$. If λ is of the order of 10^{-9} ohm^{-1} it follows that $\tau_i \geqslant \tau_d$ only if $a_0 \leq 5$ Å. This is of the order of the radius of a small ion with its solvation shell. Thus, unless ionic transport at surfaces is one or two orders of magnitude less than we have reason to believe, application of dielectric theories developed for insulating systems is never fully justified for aqueous solutions at ordinary temperatures. The same statement applies for non-spherical ions.

The Kirkwood–Shumaker protonic mechanism[13] will become important if it is the major contribution to λ, and if κ_j/κ_1 is of order unity or greater. Its order of magnitude cannot be ascertained because the mobilities of the protons among the proton acceptor sites is not known from either

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experiment or theory. The fluctuation polarization treatment[13] for the low frequency dielectric increments contains the implicit assumption that the internal field is the one appropriate to insulating systems, as the solvent conductivity does not appear anywhere in the equations. To determine the possible importance of the protonic contribution, experimental results are needed on solutions of large and rigid proteins, for which the orientational and ionic contributions may be separable through analysis of dispersion data, preferably under isoionic conditions, to prevent masking by the counterion contribution.

Aqueous Suspensions, Emulsions.—The positive dielectric increments observed in aqueous colloidal suspensions—for example, V_2O_5,[50] silver halide and arsenic sulfide sols,[51] clays,[8] glass particles,[8] and polystyrene spheres[16]—can be interpreted with the aid of the above equations. Perhaps one of the best systems for experimental test of the theory is the polystyrene system, as uniform spheres of various sizes are available,[52] and polystyrene is known to be a non-hygroscopic material of low dielectric constant and negligible volume conductivity. The density of the particles[53] is close to the bulk density of polystyrene[54] so there would appear to be no complications arising from porosity. A study of Schwan and Maczu[16] on concentrated suspensions (~30%) is reported[55] to give reasonable values for the surface conductivity. At the high concentrations employed, particle interactions are important, and there is the possibility of anisometric aggregates. Also, the composition and concentration of the surfactant stabilizer and electrolyte were not controlled. Therefore the data cannot be employed as a definitive test of this theory, but controlled experiments of the same type on more dilute systems might be.

Wide distributions of relaxation times and the high dielectric increments observed with clay and V_2O_5 sols[8] are as expected for polydisperse systems. Dispersion studies of more monodisperse systems of this type might provide an interesting quantitative test of the theory, and yield important information regarding the distribution and properties of counterions.

Emulsions of a conducting liquid in a relatively insulating medium are expected to give positive dielectric increments at all frequencies if the dielectric constant of the suspended phase is higher than the suspending medium, and at frequencies below the Maxwell–Wagner dispersion region when the dielectric constant is lower. The effects of surface conductivity will be masked by internal volume conductivity, if $\kappa_2^0 >> 2\lambda a_0$. When the suspending medium is the better volume conductor, adsorption of ions (*e.g.*, a dissociating surfactant) at the interface can give rise to surface conductivities such that $\kappa_s' = 2\lambda/a_0 > \kappa_1$. Then positive dielectric increments are expected at frequencies below the dispersion region, as illustrated for $p = 1$ in Fig. 4, and the properties of adsorbed ions may be investigated with the aid of dielectric dispersion studies.

The low frequency conductivities of suspensions of glass spheres have been studied by Fricke and Curtis.[56] The computed values for the surface conductance were of the order of 10^{-9} ohms^{-1}. Unfortunately, the concentrations of the suspensions they employed were so great ($\delta_2 = 0.26$ to 0.33) that the conductivity increments were negative, particle interactions were high, and their application of Rayleigh's equation probably is not justified. A similar study on dilute suspensions, preferably at lower electrolyte concentrations and over a range of frequencies, is needed to adequately test this model.

Colloidal Electrolytes; Synthetic Polyelectrolytes.—The dielectric properties of micellar systems[8,57,58] may be interpretable in part in terms of the present model, in preference to the Debye–Falkenhagen treatment, when large or very anisometric micelles are formed. When the micelles are small, the ion atmosphere may be thick compared to all of the particle dimensions, and the equations given here would provide only a crude approximation—perhaps no better than the Debye–Falkenhagen treatment. Adding the ion atmosphere thickness to values for the semi-axes will constitute an approximate correction. When this is done, the volume average value of the dielectric constant over particle and solvent within the equivalent ellipsoid should be used in place of the particle dielectric constant. Because there is controversy regarding the shape of micelles, and the available data are limited in scope, no attempt will be made to interpret the experimental results in a quantitative way. Dielectric and conductivity dispersion studies might be of aid in answering some of the remaining structural questions.

The dielectric properties of sodium carboxymethylcellulose were investigated by Allgen and Roswall[59] who noted that the behavior was similar to that of other polyelectrolytes, such as desoxyribonucleate, nucleohistone, and sodium hyaluronate. A study of aqueous solutions of poly-4-vinyl-*n*-butylpyridinium bromide was made by Dintzis, Oncley and Fuoss.[60] They recognized qualitatively that the high dielectric increments observed probably were associated with counterion migration, and noted that the shape of the curves suggested two dispersion regions. Both observations are consistent with the present model. They recognized a similarity to both the Debye–Falkenhagen model (diffuse ion atmosphere about a point ion) and the Maxwell–Wagner model (suspension of particles with volume conductivity), and remarked, "Whether the observed increments in

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dielectric constant are primarily due to a Maxwell–Wagner dispersion or primarily a Debye–Falkenhagen effect, or quite possibly a combination of the two, cannot be decided on the basis of the facts available at present," and that "we must not exclude the possibility that some hitherto unsuspected effect is operating in these systems." The present view is that both the Debye–Falkenhagen and the Maxwell–Wagner treatments represent approximations incorporating the same fundamental process, ion transport, with different boundary conditions. The present treatment incorporates still another set of boundary conditions which appears more acceptable for polyelectrolytes than the Debye–Falkenhagen model, because it explains many features related to size and structure. It appears more realistic than the Maxwell–Wagner model, which requires high internal volume conductivities, as these seem unreasonable for substances with compact structures.

Anisotropy of Conductivity and Dielectric Constant.—The predictions of this treatment for the effects of orientation on the dielectric increments and conductivities already have been discussed, and it was pointed out that the observations of Jacobson and co-workers on the dielectric increments in partially oriented systems are easily explained. The qualitative features of the observations of Schindewolf,[61] Heckmann[62] and Eigen and Schwarz.[63] on the anisotropy of the low-frequency conductivity in oriented polyphosphate and soap solutions are as expected for anisometric macromolecules or micelles (*cf.* Table III and IV). According to this theory, the effect should exist even when the contribution of the polyions themselves to the transport process is negligible. Schwarz[64,65] has developed a theory in which the effect is attributed solely to the lower frictional coefficient of the extended polyion along its longer axis, as compared to the perpendicular axis. In terms of the present model, the polyion mobility may be negligible, and the anisotropy effect will still exist, because of the depolarizing factor in the equations for conductivity. Also, on the present model, it will depend strongly on the mobility and the degree of binding of the counterions. Furthermore, it seems clear that the anisotropy of the polyion mobility will involve an important consideration in addition to the change in frictional coefficient, namely, the change of local field at the polyion with orientation. For the present model at low frequencies, and for orientation of the j-axes along the field

$$E_j(\text{int}) = E_j(\text{ext})/[1 + (\kappa_j/\kappa_1 - 1)A_j]$$

where E_j(int) and E_j(ext) are the internal and external field intensities along the j-axis. Referring to Table I, it is seen that at high p-values, A_a becomes very small, A_b approaches 0.5, and κ_a' and κ_b' are of the same order of magnitude. Thus, when $\kappa_j/\kappa_1 > 1$, the local field effect may far exceed the frictional coefficient consideration, in determining the anisotropy of the mobility of the polyion, defined in the customary way in terms of the external field. This will make the anisotropy of the polyion contribution even greater than supposed in Schwarz's treatment, but it does not follow that the polyion contribution will exceed the contributions from the relatively mobile counterions which are computed on the present model. A complete theory must include both effects. Direct observations of the polyion mobility as a function of electric field (or degree of orientation) will probably be required to evaluate contributions of polyions and other ions separately. The electric field orientation problem was treated independently by Schwarz[65] and O'Konski and Haltner[23] in terms of the limiting model in which the particles are considered very good conductors relative to the medium. In the latter study the absolute magnitude of the Kerr constant was studied in a carefully chosen model system,[23] and it was found to be significantly below the value calculated for the limiting case. The theory has been extended to the present model and will be presented elsewhere.[22]

Concentrated Systems: Biological Tissues, Polycrystalline Powders, Soils.—The above discussion on the origin and the magnitude of the surface conductivity effect, and its dependence upon particle size and shape is of interest in the further interpretation of the dielectric properties of tissues and cell suspensions, which have been reviewed recently by Schwan.[16]

As a consequence of a high localized conductivity in interfacial regions, special effects may be produced in biological systems by application of short bursts of intense high-frequency radiation, *e.g.*, lightning strokes or high-frequency pulses. At frequencies below the dipolar dispersion region of water, the temperature rise in the ion atmosphere of polyelectrolytic components, *e.g.*, the desoxyribonucleic acid in genes, would be expected to exceed the average temperature rise of the system until the heat can be carried away from the surface by thermal conduction. This may give rise to much greater biochemical changes in pulsed fields than expected on the basis of the mean temperature rise of the system.

The high values of the low frequency dielectric constants and the dielectric dispersion observed in crystalline powders containing very small amounts of adsorbed water vapor are very probably the result of surface conductivity produced by mobile ions in adsorbed films of water.[66,67] The presence of mobile ions on the surfaces of anisometric particles of clay probably is responsible for the large low frequency dielectric constants and the broad region of dielectric dispersion observed in soils and for the variations observed with soil type and moisture content.[68–70]

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Protein Hydration from Dielectric Studies.—Interpretation of the high frequency dielectric decrements of protein solutions in terms of a solvated model having a low dielectric constant[71,72] appears to be a sound approach to the problem for compact molecules, providing the data can be obtained in a region where the loss contributions from ionic processes make only small contributions to the field distribution ($\epsilon_j'' << \epsilon_j$). This will be so if $\tau_i > 1/\omega$, where ω is the angular frequency at which the decrement is observed. Wherever information regarding molecular asymmetry is available, the dielectric decrement appropriate to the macromolecular shape may be computed.[70] From the values of $\Delta\epsilon_h/\epsilon_1\delta_2$ of Table VI, it is clear that the calculated hydration will depend upon the molecular shape, the variation being greater in going from spheres to oblate ellipsoids than in going from spheres to prolate ellipsoids.

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COEFFICIENTS OF EVAPORATION AND CONDENSATION

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A model is developed for crystal growth and evaporation which invokes the Kossel concept of a crystal surface being composed of low index facets, of low binding energy for adsorbed atoms, separated by ledges of higher binding energy. The treatment indicates that surface diffusion kinetics are important in most instances of crystal growth and evaporation. Prior kinetic treatments, such as those involving limitations of crystal growth because of entropy effects or diffusion in the vapor phase, are modified to include surface diffusion effects. Values for evaporation and condensation coefficient predicted by the kinetic treatment are compared with experimentally determined values.

Introduction

Since the relation between the vapor pressure of a substance and the flux of the substance leaving a surface in equilibrium with the vapor was developed by Hertz,[1] Knudsen,[2] and Langmuir[3] there has been a vast research effort to investigate the quantity α_j ambiguously called the condensation coefficient, the evaporation coefficient, or the accommodation coefficient in the expression

$$J_i = \pm \alpha_j(p_e - p)/(2\pi mkT)^{1/2} \qquad (1)$$

where the plus sign is used for evaporation, the minus for condensation, $J_i = J_v$ is the net vaporization flux in molecules per cm.^{2} per second, $J_i = J_c$ is the net condensation flux, $j = 1, 2, 3, \cdots$, dependent on the process being considered, p_e = the equilibrium pressure, p = the actual vapor pressure, m = molecular mass, and k and T have their usual meaning. In general, α_j in condensation does not equal α_j in evaporation under similar experimental conditions as was assumed by early investigators. Also, there are often several factors affecting α_j which may or may not be mutually independent. Accordingly, the following treatment of the various factors affecting J_i is presented in order that the interaction of these factors might be clarified. The various factors are considered separately, the simpler cases being treated initially and complexities introduced in sequence. In each case, the consideration of surface diffusion in the kinetics represents the new contribution of this work.

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Theory

I. Clean Crystalline Surface with a Monatomic Vapor Phase.—In this case, the internal partition functions of an atom in the crystal are vibrational and in the vapor phase are translational. Hence, rotational degrees of freedom are not activated in evaporation or condensation and, as a first approximation, it is possible to use a model of evaporation and growth in which atoms are considered to be bound by near neighbor bonds.[4,5] It has been shown by Burton, Cabrera and Frank[6] in the case of growth and by Hirth and Pound[7] in the case of evaporation, both following the general ideas of Volmer,[8] that the kinetics involve the steps: (a) adsorption or desorption to the vapor phase from the surface; (b) surface diffusion; and (c) movement into or out of a kink position in a monatomic ledge on the crystal surface, at which the equilibrium concentration of adsorbed atoms is maintained, see Fig. 1.

For ideal crystals, *i.e.*, large crystals bounded by low index planes and with no lattice imperfections other than the monatomic ledges and adsorbed atoms, the monatomic ledges arise from crystal edges in evaporation. The adsorbed atoms on the surface develop the distribution shown in Fig. 2, where it is shown that the equilibrium concentration of adsorbed atoms is maintained at

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